PYTHON ENGINEERING

EX08 - Discretization and the Euler Method

From a continuous differential equation to a step-by-step numerical algorithm

$$dh/dt = v \mid dv/dt = -g$$

Goal: understand why Euler uses h(k+1) = h(k) + v(k) dt and why the term -1/2 g dt^2 does not appear directly. We start with physical intuition and finish by connecting it to the mathematical basis of the method.

Learning Tracker

Concept	Goal
ODE / EDO	Interpret a derivative as an evolution law
Discretization	Convert continuous time into dt steps
Explicit Euler	Use the current slope to advance one step
Numerical error	Understand why the approximation is not exact
Convergence	See how reducing dt improves the solution

Key idea: the exact kinematic formula and the Euler method are not the same algorithm. The former uses integrated information for constant acceleration; Euler uses only the instantaneous slope at the beginning of each step.

1. The question that starts everything

For vertical motion with constant gravity, mechanics gives us:

$$h(t + dt) = h(t) + v(t)\, dt - 1/2\, g\, dt^2$$

$$v(t + dt) = v(t) - g\, dt$$

So it is natural to ask: if this expression is correct, why does Euler use only:

$$h(k+1) = h(k) + v(k)\, dt$$

Short answer: because Euler is not using the exact kinematic solution. It approximates the derivative dh/dt = v using a local straight line.

Two different problems

Approach	Information used	Position step
Exact kinematics	We know a = -g is constant	h_new = h + v dt - 1/2 g dt^2
Explicit Euler	We only use dh/dt = v at the current state	h_new = h + v dt

There is no contradiction: the first expression is richer because it exactly accounts for how velocity changes within the interval. Euler freezes the slope during that small step.

2. Mathematical basis: what a derivative means

The derivative of h(t) is defined as the limit of an incremental quotient:

$$dh/dt = \lim(dt \to 0) [h(t + dt) - h(t)] / dt$$

If dt is not infinitesimally small but a finite numerical step, Euler replaces the limit with an approximation:

$$dh/dt \approx [h(k+1) - h(k)] / dt$$

Because dh/dt = v, we evaluate the slope at the current state k:

$$v(k) \approx [h(k+1) - h(k)] / dt$$

Solving for h(k+1):

$$h(k+1) = h(k) + v(k)\, dt$$

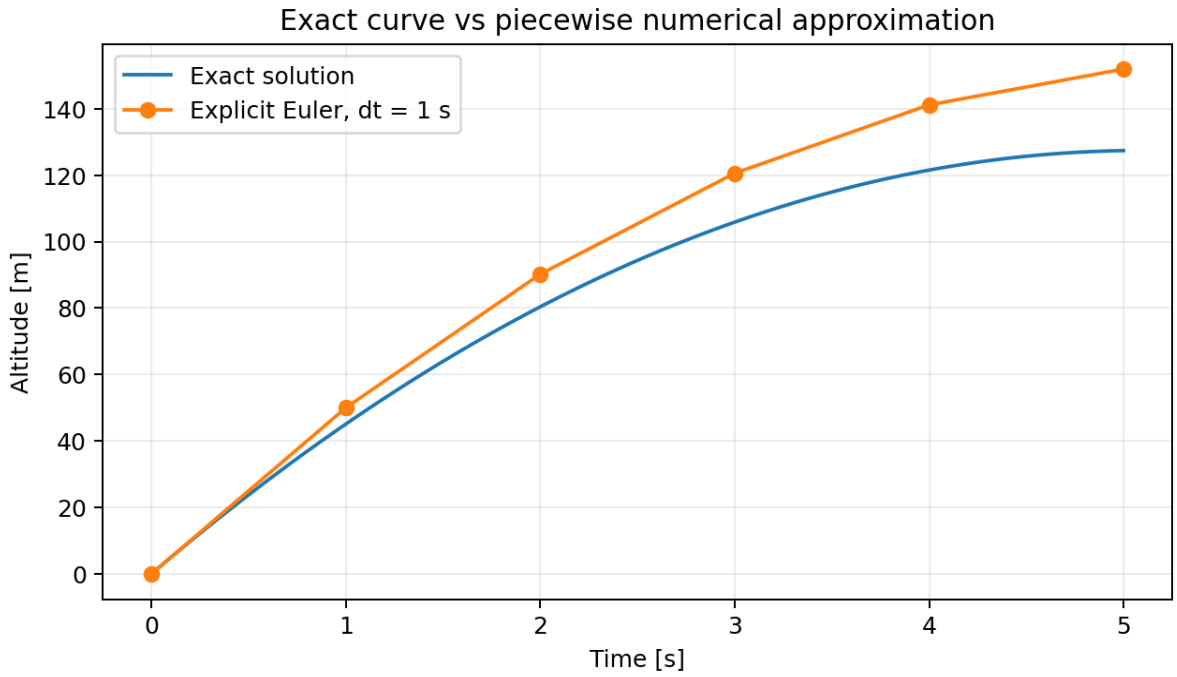
The same idea applied to dv/dt = -g gives:

$$v(k+1) = v(k) - g\, dt$$

Euler is essentially a tangent-line approximation: we move forward by dt assuming the slope remains equal to its value at the beginning of the step.

3. Geometric interpretation: curve vs tangent line

The exact trajectory h(t) is a parabola. Euler does not know the full parabola; it only knows the slope v(k) at the current instant. Therefore it builds a sequence of short straight segments.



With a large dt the segments are long and depart more from the curve. With a small dt each segment covers a region where the slope changes less, so the approximation improves.

The idea of discretization

$$t(k) = t0 + k\,dt$$

Instead of knowing the state at every possible time, we compute it only on a grid: t0, t1, t2, ... Each point is obtained from the previous one.

4. Numerical example: a single step

Take h0 = 0 m, v0 = 50 m/s, g = 9.81 m/s^2 and dt = 1 s.

Euler

$$h1 = 0 + 50(1) = 50.000 \text{ m}$$

$$v1 = 50 - 9.81(1) = 40.190 \text{ m/s}$$

Exact kinematics

$$h1 = 0 + 50(1) - 1/2(9.81)(1)^2 = 45.095 \text{ m}$$

The position difference after that step is:

$$50.000 - 45.095 = 4.905 \text{ m}$$

And 4.905 m is exactly the term Euler ignored over that interval:

$$1/2 \, g \, dt^2 = 4.905 \text{ m}$$

Important: this does not mean we should add that term to Euler. If we add it, we are changing the algorithm and exploiting the fact that acceleration is constant in this specific problem.

5. What happens when dt is reduced

The curvature term Euler does not include within one step is proportional to dt^2. For the same g:

dt [s]	1/2 g dt^2 [m]	Interpretation
1	4.905000	large
0.5	1.226250	large
0.1	0.049050	small
0.01	0.000491	very small

As dt decreases, the error made in each individual step drops rapidly. However, we perform more steps. For Euler, the local truncation error is of order dt^2, while the accumulated global error is of order dt.

$$\text{Error local} \sim O(dt^2) \mid \text{Error global} \sim O(dt)$$

In practical terms: if you roughly halve dt, for a well-behaved problem you expect Euler's global error to roughly halve.

Convergencia

$$dt \to 0 \Rightarrow \text{solucion numerica} \to \text{solucion exacta}$$

Convergence does not mean that a finite dt gives perfect accuracy. It means the method approaches the correct solution as the discretization is sufficiently refined.

6. From the ODE system to the algorithm

$$\text{State } x = [h, v]$$

$$dx/dt = f(x,t) = [v, -g]$$

Euler can be written in vector form:

$$x(k+1) = x(k) + f(x(k), t(k))\, dt$$

For our case:

```
new_h = h + v * dt
new_v = v - g * dt

h = new_h
v = new_v
t = t + dt
```

First compute the entire new state from old-state values. Only then replace h and v. This implements explicit Euler clearly and avoids mixing states k and k+1.

Why use new_h and new_v

If you update v before computing h, then h would use v(k+1) instead of v(k). That corresponds to a different scheme. It is not necessarily useless, but it is no longer the explicit Euler method we want to study.

7. Euler is not designed to exploit a special formula

For constant gravity we can integrate exactly. But interesting engineering problems rarely keep acceleration this simple. For example, with aerodynamic drag:

$$dv/dt = -g - [rho(h)\ C_D\ A / (2m)]\ v\ |v|$$

Now rho depends on altitude, drag depends on v^2, and C_D may depend on Mach. There is no longer a single elementary term such as -1/2 g dt^2 that we can add to solve everything.

That is the real pedagogical reason to study Euler: it is a general method. It may be inaccurate, but it teaches the basic pattern that we will later improve with RK2, RK4 and solve_ivp.

Method progression

Method	Main idea	Global order
Euler	1 slope per step	O(dt)
RK2	Combines 2 slopes	O(dt^2)
RK4	Combines 4 slopes	O(dt^4)
solve_ivp	Professional adaptive integrators	depends on method

8. Recommended EX08 experiment

First implement pure Euler, without the -1/2 g dt^2 term, and run the same launch with different time steps:

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dt = 1.0
dt = 0.5
dt = 0.1
dt = 0.01
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For each case compare:

• total flight time

• maximum altitude

• final velocity

• number of iterations

against the analytical references:

$$t_{flight} = 2 v_0 / g$$

$$h_{max} = v_0^2 / (2g)$$

Questions you should be able to answer at the end

• Why does Euler not include -1/2 g dt^2?

• What information does Euler use to advance?

• What does discretizing time mean?

• Why does reducing dt improve the solution?

• Why is Euler still useful if an exact formula exists for this case?

Sentence to remember: Euler does not try to know the full curve; it advances along the local tangent, one step at a time.